Sleep, circadian rhythm, and recovery

Week 2 Lecture 2 · Looksmaxxing

Evidence-based appearance improvement · Autumn 2026

What we'll cover today

- **The evidence on sleep and facial appearance** , a blinded study, not a wellness claim
- **What happens physiologically** during sleep deprivation: skin, hormones, and mood
- **The AASM minimum:** what the consensus says and why it matters
- **Circadian rhythm and light hygiene:** the levers you can actually pull
- **Caffeine half-life** and its effect on sleep architecture
- **Habit-stacking** a consistent sleep schedule

What sleep deprivation does to your face

The Sundelin et al. study (2017)

Not a wellness blog. A blinded experimental study.

- **Study design:** Royal Society Open Science, vol. 4, article 160918
- **Method:** participants restricted to five hours per night for two consecutive nights, then photographed; a separate group of blinded observers rated the photos
- **What raters saw:** sleep-deprived faces were rated as paler, less healthy, sadder, and less attractive
- **Effect size:** statistically significant drops in attractiveness, health, and trustworthiness ratings

(Sundelin, Lekander, Sorjonen, Axelsson, 2017)

What the raters noticed

Blinded observers reported specific visual cues:

- Paler skin color
- More hanging eyelids
- More swollen eyes and darker under-eye circles
- More drooping corners of the mouth
- More visible wrinkles and fine lines

These were two nights of five-hour sleep. Not chronic deprivation. Two nights.

The physiological mechanism

Sleep is not passive downtime. It is the period when your body runs maintenance:

- **Collagen synthesis** peaks during slow-wave sleep
- **Growth hormone** is secreted primarily during the first hours of sleep
- **Cortisol** drops overnight; sleep deprivation keeps it elevated
- **Trans-epidermal water loss** increases with disrupted sleep, reducing skin barrier integrity

Cutting sleep is not just cutting rest. It is cutting the maintenance window.

The consensus recommendation

What the AASM says

The American Academy of Sleep Medicine and Sleep Research Society issued a joint consensus statement in 2015:

> "Adults should sleep 7 or more hours per night on a regular basis to promote optimal health."

This was a panel of 15 sleep medicine experts, 12 months of evidence review.

The statement documents specific adverse outcomes for sleeping fewer than seven hours: weight gain, obesity, diabetes, hypertension, and cardiovascular disease.

(AASM & Sleep Research Society, 2015)

Sleep deprivation and body composition

Cutting sleep below seven hours affects body composition through two pathways:

1. **Ghrelin rises, leptin falls** , appetite regulation shifts toward increased hunger, especially for calorie-dense foods
2. **Insulin sensitivity decreases** , even one week of five-to-six hour nights produces measurable insulin resistance in healthy adults

Both effects work directly against body composition goals. You cannot strength-train your way out of a chronic cortisol and insulin sensitivity problem driven by insufficient sleep.

Circadian rhythm and light

What circadian rhythm actually is

Your circadian clock is a roughly 24-hour biological oscillator driven by light signals hitting the retina.

It controls:

- When melatonin is secreted (onset: 2-3 hours before typical sleep time)
- Core body temperature rhythm (drops overnight, rises before waking)
- Cortisol pulse (peaks within one hour of natural wake time)

The clock drifts without regular light input. Irregular sleep timing means irregular cortisol, irregular melatonin, and disrupted skin recovery cycles.

Light hygiene in practice

The circadian clock is reset by:

Light in the morning: bright light within one hour of waking advances the clock toward an earlier sleep onset. Outdoor light is 10,000+ lux. Indoor overhead light is 100-300 lux. The difference matters.

Light in the evening: blue-wavelength light from screens and overhead LEDs suppresses melatonin onset. Dimming overhead lights and using night-mode settings after 9pm reduces this suppression.

This is not complicated. It is two actions: more light in the morning, less in the evening.

Caffeine half-life

Caffeine has an average half-life of five to seven hours in healthy adults.

A 200mg coffee at 3pm:

```
3pm:  200mg caffeine
8pm:  ~100mg remaining (one half-life at 5h)
1am:  ~50mg remaining
6am:  ~25mg remaining
```

"I can drink coffee at 9pm and sleep fine" is almost certainly false, or your sleep architecture is disrupted even if you fall asleep. Caffeine blocks adenosine receptors without clearing the adenosine, so sleep pressure is suppressed but the biology of the maintenance window is still degraded.

The evidence-based cutoff: no caffeine after 2pm for most people.

The habit machine

Why sleep timing is a habit problem

You already know what good sleep looks like. The gap is not information.

James Clear (2021): every habit has a cue-craving-response-reward loop. Consistent bedtime is not self-discipline. It is a habit that fires reliably because it is anchored to an existing behavior.

The formula: “After I [CURRENT HABIT], I will [NEW SLEEP BEHAVIOR].”

Example: “After I plug in my phone to charge, I will turn off the overhead light and not pick the phone up again.”

Building the stack

A sleep habit-stack has three components:

1. **Anchor event** , something you already do every night at roughly the same time
2. **Friction reduction** , make the bad behavior harder (phone in another room, app timer, no caffeine after 2pm)
3. **Reward signal** , brief celebration or acknowledgment when you complete the stack, not just at the outcome

The stack does not need to be elaborate. It needs to fire automatically. Elaborate routines fail when life varies. A two-step stack survives.

Take-aways

1. Two nights of five-hour sleep produces measurable, rated drops in facial attractiveness, skin pallor, and health perception , a blinded experimental finding, not a wellness claim (Sundelin et al., 2017).
2. The AASM consensus sets seven or more hours per night as the adult minimum for health. Below that, metabolic and appearance consequences accumulate (AASM, 2015).
3. Circadian rhythm is driven by light timing. Morning bright light and reduced evening blue light are the two evidence-based levers.
4. Caffeine has a five-to-seven hour half-life. A 3pm coffee still has active concentration at midnight.
5. Consistent sleep timing is a habit problem, not a willpower problem. A two-step anchor-based habit stack is more durable than any elaborate routine (Clear, 2021).

> "Two days of restricted sleep (5h/night) significantly decreased perceived attractiveness, health, and trustworthiness... Faces of sleep-deprived individuals were perceived as paler, less healthy, and sadder than well-rested faces."

- Sundelin, Lekander, Sorjonen, Axelsson · Royal Society Open Science (2017)

What's next

- **Section (this week):** Set up your photo station, take week-zero photos, build your habit-stack on paper
- **Reading:** Week 2 reading, baseline, sleep, and the habit machine
- **Week 3:** Skincare , the highest-leverage daily intervention. Your habit-stack from this section is the delivery mechanism for the Week 3 routine.
- **No assignment due this week**